

Condensate Partition Coefficient Pipeline

Combined Report

User Manual • Methods • Research Timeline • JABr Reference Data

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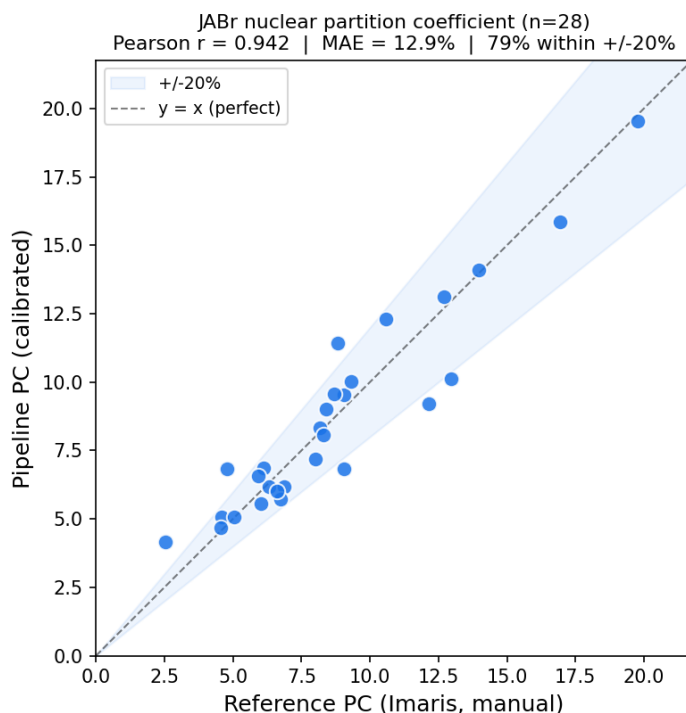
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1 Part I — User Manual

This pipeline replaces the manual Imaris workflow for measuring the partition coefficient (PC) of nuclear biomolecular condensates. Given a two-channel confocal Z-stack (channel 0 = nuclei, channel 1 = condensate), it segments nuclei and condensates in 3D, computes the PC, and reports a calibrated value on the manual reference scale. It runs from a point-and-click graphical interface or the command line.



Validation on JABr: calibrated pipeline PC vs the manual Imaris reference.

1.1 Installation (one time)

Install Python 3.10 or 3.11, then install PyTorch for your machine (CPU or your CUDA version) from pytorch.org. Example for CUDA 12.1:

```
pip install torch --index-url
https://download.pytorch.org/whl/cu121
```

Then install the remaining dependencies:

```
pip install -r requirements.txt
```

On first run, Cellpose downloads its model weights automatically (one-time ~100 MB). A GPU is recommended (~10–15 s per cell); CPU works but is slower.

1.2 Preparing image files

The pipeline accepts either one multi-channel TIF (both channels in one file) or two separate single-channel TIFs. **Channel order matters:** channel 0 must be nuclei, channel 1 the condensate. Most axis orderings (ZCYX, CZYX, ...) are auto-detected; if results look wrong, swapped channels are the most common cause.

1.3 Running with the graphical interface

Launch with `python run_gui.py`. A window with **Single File** and **Batch** tabs opens.

1. On the **Single File** tab, choose the input mode and browse to your data.
2. Choose an output folder (optional).
3. In **Settings**, set **Construct = JABr**; leave all other settings at their defaults.
4. Click **Run Pipeline**. Progress streams in the Output Log; the status reads **Done** when finished.
5. Open your output folder for the masks, tables, and `results.png`.

1.4 Settings

For routine JABr analysis you only need to set **Construct**; the rest have sensible defaults.

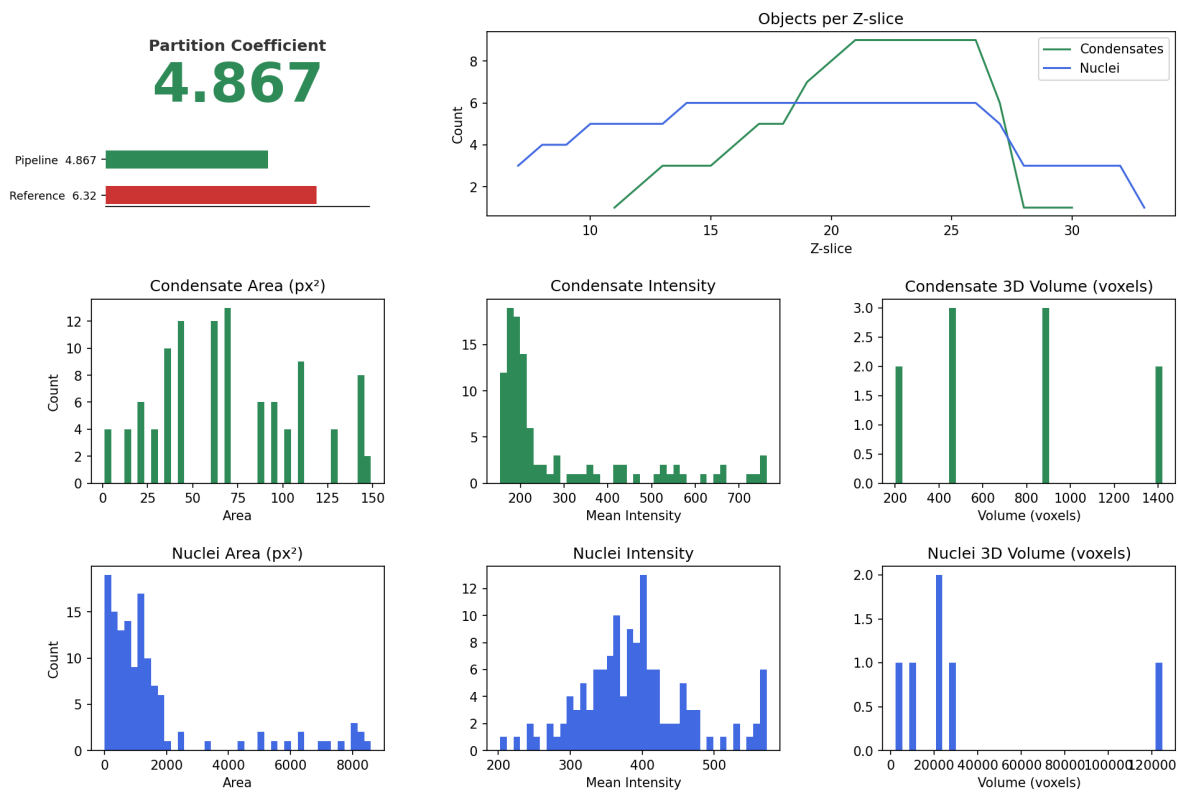
Setting	Default	What it does
Construct	(none)	Pick JABr for the validated workflow. Auto-selects the <code>blob_log</code> detector and the JABr calibration so the reported PC is on the manual scale.
Condensate detector	auto	auto routes JABr to <code>blob_log</code> , else Cellpose. Normally left on auto.
Top-X% brightest	75	Defines condensed-phase density; trims the dim mask halo to match the manual reference. Leave at 75.
Nuclei cell-prob.	-2.0	How aggressively nuclei are detected. -2.0 suits the lab's nuclear stain.
blob_log threshold	0.03	Condensate spot sensitivity (used when detector = <code>blob_log</code>). Leave at 0.03 for JABr.
Disable GPU	off	Tick on a laptop with no NVIDIA GPU.

The one rule for routine use: set Construct = JABr and run.

1.5 Reading the outputs

Every run writes `summary.csv` (headline numbers including the calibrated PC), `results.png`, `condensate_masks.tif` and `nuclei_masks.tif` (3D labeled masks), and per-object/per-slice CSVs. For the bundled sample `JABr_Sample2_5_3.tif`, a real run produces nuclear PC raw 4.867 calibrated

to 4.813 (manual reference 4.558, +5.6%). Report the calibrated nuclear PC, and always glance at `condensate_masks.tif` over the raw channel in Fiji/ImageJ to catch the rare misfire.



results.png from the bundled-sample run.

1.6 Command line

```
python pipeline.py --roi sample_data/JABr_Sample2_5_3.tif
--construct JABr --output my_output
```

Separate channels: `--nuc nuclei.tif --cond condensate.tif`. Physical volumes in μm^3 : add `--voxel-xy 0.065 --voxel-z 0.3`. Force CPU: `--no-gpu`. For many files (and optional comparison vs a manual reference CSV), use the GUI **Batch** tab.

1.7 Troubleshooting

Symptom	Likely cause / fix
PC wildly off / nuclei empty	Channels swapped: confirm ch0 = nuclei, ch1 = condensate.
“0 condensates detected”	Not JABr, or very dim signal. Lower blob_log threshold (e.g. 0.02).
Very slow (minutes/cell)	Running on CPU. Install a CUDA PyTorch build; ensure Disable GPU is unchecked.
CUDA out of memory	Tick Disable GPU, or close other GPU programs.
One cell’s calibrated PC off	Calibration is population-level; cells vary ($\pm 20\%$ for $\sim 80\%$ of cells).

2 Part II — Methods

2.1 The quantity

The partition coefficient is $PC = \text{condensed-phase density} / \text{dilute-phase density}$ (Fabrini et al.). All intensities are background-subtracted, where the background B is the minimum voxel intensity across the field of view (camera offset).

- **Condensed-phase density** — mean of $\text{clip}(\text{intensity} - B, 0)$ over the brightest 75% of voxels inside both a condensate mask and a nucleus. Automated masks include a dim halo a manual tracing would exclude; trimming the bottom 25% recovers densities consistent with the manual reference.
- **Dilute-phase density** — mean of the 50 lowest-intensity valid $10 \times 10 \times 10$ -voxel patches inside the nuclear dilute region. Averaging many low patches approximates a manual quiet-region selection and is far more stable than a single random patch.

2.2 Pipeline steps

Step	Method	Why
Denoise	Cellpose 3 <code>denoise_cyto3</code> per slice	Removes shot noise; PC intensities are read from the raw stack.
Segment nuclei	Cellpose 3 <code>cyto3</code> , <code>do_3D</code>	Native 3D nuclear segmentation.
Clean nuclei	Connected-component re-label; drop < 1000 -voxel fragments	Cellpose over-splits large nuclei; re-labeling restores whole nuclei.
Fill voids	Per-nucleus <code>binary_fill_holes</code> (3D + 2D)	Condensates carve donut holes in the nuclear stain exactly where nuclear condensates sit; filling makes them count as intra-nuclear.
Detect condensates	<code>blob_log</code> (LoG), σ 1.5–6.0, threshold 0.03	Classical, parameter-light detector matched to roughly-spherical condensates; each rendered as a sphere of radius $\sigma\sqrt{3}$.
Intra-nuclear gate	Keep a blob if $\geq 50\%$ of its sphere lies inside a nucleus	Replaces an older single-centroid test that mis-handled boundary blobs and crowded fields.
Calibrate	Per-construct linear (JABr) / isotonic map	Standardizes the automated PC onto the manual Imaris scale.

2.3 Why there is a calibration

The automated and manual workflows define the condensed and dilute regions slightly differently: the automated condensed region runs brighter (brightest mask voxels) and the automated dilute region runs dimmer (quietest sampled patches) than a hand-drawn Imaris region. Both push the ratio the same way, so the raw automated PC is systematically about $3\times$ the manual reference — but consistently so ($r = 0.94$). A single linear map converts it back:

$$\text{calibrated_PC} = 0.3642 * \text{raw_PC} + 3.0405 \quad (\text{JABr})$$

This is cross-method standardization, not a fudge factor. A linear map is rank-preserving, so calibration cannot improve correlation — it only removes the scale/offset bias. The correlation r is therefore the true accuracy ceiling, and calibration is meaningful only where r is already high.

2.4 Scope and generalization

The pipeline is validated and production-ready for JABr. It is construct-specific: the `blob_log` parameters encode JABr’s condensate size and brightness and do not transfer unchanged. Cross-construct correlation vs Imaris:

Construct	Pearson r	Status
JABr	0.942	Production
GABr	0.678	Mediocre; isotonic calibration, use with caution
AABr	0.122	Not usable as-is
AwtBr	−0.116	Not usable as-is
GwtBr	−0.341	Not usable as-is

A leave-one-construct-out test (a single learned Cellpose detector, evaluated on a construct it never trained on) confirmed that one model does not generalize zero-shot: it fails by under-detection, staying silent on most cells of an unseen construct. The route to extend the tool is per-construct calibration where correlation is already acceptable, or few-shot retraining with a handful of labeled ROIs of the new construct — far cheaper than a manual parameter sweep.

3 Part III — Research Timeline

A chronological record of how the pipeline was built. Dates are from the project’s version-control history.

3.1 Winter 2026 — Foundation

Date	Milestone
2026-03-17	Initial clean pipeline established.
2026-03-24	Winter 2026 report finalized; baseline segmentation + PC concept.

3.2 April 2026 — Segmentation survey & the PC formula

Date	Milestone
2026-04-22	Spring segmentation-model survey; 4 candidate models compared.
2026-04-25	Background-subtracted PC formula adopted across all scripts.
2026-04-30	PC now matches reference: connected-component nuclei fix + lowest-patch dilute density; 50-patch stability fix.

3.3 May 2026 — Calibration, GUI, generalization

Date	Milestone
2026-05-06	Batch cross-reference pipeline; top-75% condensate-density calibration.
2026-05-07	GUI built; laptop (CPU) validation; PI meeting.
2026-05-14	Crowded-field debugging; max-overlap-nucleus target-cell heuristic.
2026-05-16	First fine-tuned Cellpose model; 5-construct accuracy sweep; per-construct calibration.
2026-05-18	M187 research poster (PC = 6.297 vs 6.32 reference).
2026-05-28	blob_log detector wired into the pipeline; JABr calibration refit; GUI gains construct/detector controls; cytoplasmic PC added.

3.4 June 2026 — Production hardening & honest scope

Date	Milestone
2026-06-10	Discussion section; nuclei void-filling so condensates in donut holes count toward the PC.
2026-06-11	Sphere-overlap intra-nuclear gate promoted to production: JABr r 0.926 to 0.942, MAE 14.6% to 13.7%.
2026-06-12	Leave-one-construct-out experiment: a single learned detector does not generalize zero-shot. Few-shot path mapped.
2026-06-13	Clean public release: repository, instruction manual, methods writeup, and this combined report.

3.5 Where it stands

Validated, production-ready for JABr ($r = 0.942$, MAE 12.9%, 79% of cells within $\pm 20\%$, $n = 28$). Delivered: a CLI, a GUI for non-coding lab use, per-construct calibration, and full documentation. Known boundary, with a plan: the detector is JABr-specific; extending it needs per-construct calibration or few-shot retraining.

4 Part IV — JABr Reference Data

Source: Box → Condensate Volume Quantification → JABr. The manual Imaris measurements (nuclear and cytoplasmic) are the ground truth against which the pipeline is validated; the pipeline results table is the production `blob_log V4` output (sphere-overlap gate, linear calibration).

4.1 Manual reference — Nuclear (n = 30)

File	Condensate Density	Dilute Density	Partition Coefficient
Sample1_1_1	390.42	81.67	4.780
Sample1_1_2	827.87	91.60	9.038
Sample1_1_3	1516.55	76.69	19.774
Sample1_1_4	793.32	74.96	10.584
Sample1_2_1	619.05	77.16	8.023
Sample1_2_2	339.36	73.78	4.599
Sample1_2_3	162.87	63.99	2.545
Sample1_4_1	540.60	88.20	6.129
Sample1_4_2	368.83	73.05	5.049
Sample1_4_3	1338.28	95.84	13.963
Sample1_4_4	584.31	62.68	9.322
Sample2_5_1	658.30	104.09	6.324
Sample2_5_2	483.19	70.30	6.873
Sample2_5_3	331.24	72.67	4.558
Sample2_5_4	415.30	68.99	6.020
Sample2_5_5	497.00	73.88	6.728
Sample2_5_6	894.09	73.58	12.151
Sample3_3_1	534.27	65.28	8.185
Sample3_3_2	633.55	70.04	9.045
Sample3_3_3	788.77	89.38	8.825
Sample3_3_4	481.04	72.90	6.599
Sample3_3_5	627.55	74.59	8.413
Sample3_3_6	1365.81	107.64	12.689
Sample3_3_7	719.19	82.84	8.681
Sample3_3_8	607.50	73.24	8.295
Sample3_3_9	455.50	76.94	5.920
Sample3_3_10	205.48	56.92	3.610
Sample3_3_11	1361.65	80.33	16.951
Sample3_3_14	907.64	69.99	12.969
Sample3_3_15	426.20	66.50	6.409

4.2 Manual reference — Cytoplasmic (n = 30)

File	Condensate Density	Dilute Density	Partition Coefficient
Sample1_1_1	614.56	79.99	7.683
Sample1_1_2	972.96	160.44	6.064
Sample1_1_3	265.66	75.78	3.505
Sample1_1_4	567.86	73.77	7.698
Sample1_2_1	325.21	66.05	4.924
Sample1_2_2	174.05	71.28	2.442
Sample1_2_3	99.17	56.51	1.755
Sample1_4_1	235.26	59.68	3.942
Sample1_4_2	243.07	74.47	3.264
Sample1_4_3	234.73	62.48	3.757
Sample1_4_4	283.53	57.32	4.946
Sample2_5_1	411.70	85.14	4.836
Sample2_5_2	212.39	61.12	3.475
Sample2_5_3	289.37	70.18	4.123
Sample2_5_4	298.08	73.13	4.076
Sample2_5_5	259.89	62.52	4.157
Sample2_5_6	388.44	67.07	5.791
Sample3_3_1	280.76	62.75	4.475
Sample3_3_2	154.32	68.10	2.266
Sample3_3_3	468.87	68.28	6.867
Sample3_3_4	237.35	69.67	3.407
Sample3_3_5	787.46	66.95	11.761
Sample3_3_6	226.58	82.30	2.753
Sample3_3_7	317.80	63.43	5.010
Sample3_3_8	411.18	65.28	6.298
Sample3_3_9	233.25	71.88	3.245
Sample3_3_10	1351.58	93.56	14.446
Sample3_3_11	658.92	92.78	7.102
Sample3_3_14	212.48	66.59	3.191
Sample3_3_15	110.00	67.56	1.628

4.3 Pipeline results — Nuclear (blob_log V4, sphere-overlap gate)

Detector blob_log (threshold 0.03, σ 1.5–6.0); nuclei via Cellpose cyto3 with void-filling; intra-nuclear gate $\geq 50\%$ sphere overlap; calibration $0.3642 \cdot \text{raw} + 3.0405$. $n = 28$ (Sample3_3_10 and Sample3_3_15 find 0 intra-nuclear condensates and are excluded).

File	Cond Dens.	Dilute Dens.	Bg	Raw PC	Cal PC	Ref PC	lerrl %
Sample1_1_1	588.5	56.6	89	10.39	6.82	4.78	42.8%
Sample1_1_2	1151.2	64.7	93	17.80	9.52	9.04	5.4%
Sample1_1_3	2168.3	47.9	99	45.28	19.53	19.77	1.2%

File	Cond Dens.	Dilute Dens.	Bg	Raw PC	Cal PC	Ref PC	err %
Sample1_1_4	1369.7	53.8	92	25.45	12.31	10.58	16.3%
Sample1_2_1	632.2	55.5	92	11.40	7.19	8.02	10.4%
Sample1_2_2	350.7	63.1	87	5.55	5.06	4.60	10.1%
Sample1_2_3	162.1	53.2	95	3.05	4.15	2.55	63.0%
Sample1_4_1	573.6	54.8	92	10.47	6.85	6.13	11.8%
Sample1_4_2	364.0	65.6	87	5.55	5.06	5.05	0.3%
Sample1_4_3	1765.7	58.1	88	30.39	14.11	13.96	1.0%
Sample1_4_4	957.1	49.9	91	19.19	10.03	9.32	7.6%
Sample2_5_1	606.5	70.1	76	8.65	6.19	6.32	2.1%
Sample2_5_2	544.3	63.3	84	8.60	6.17	6.87	10.2%
Sample2_5_3	319.5	71.4	79	4.47	4.67	4.56	2.4%
Sample2_5_4	414.2	60.1	86	6.89	5.55	6.02	7.8%
Sample2_5_5	414.6	56.4	84	7.35	5.72	6.73	15.0%
Sample2_5_6	1083.0	64.0	82	16.92	9.20	12.15	24.3%
Sample3_3_1	797.3	54.9	88	14.53	8.33	8.18	1.8%
Sample3_3_11	1970.5	56.0	84	35.20	15.86	16.95	6.5%
Sample3_3_14	1149.6	59.2	85	19.42	10.11	12.97	22.0%
Sample3_3_2	658.0	63.4	89	10.39	6.82	9.05	24.6%
Sample3_3_3	1361.7	59.2	88	23.01	11.42	8.82	29.4%
Sample3_3_4	453.8	55.8	89	8.13	6.00	6.60	9.1%
Sample3_3_5	876.9	53.3	86	16.44	9.03	8.41	7.3%
Sample3_3_6	1610.5	58.2	88	27.68	13.12	12.69	3.4%
Sample3_3_7	1018.8	56.8	89	17.93	9.57	8.68	10.2%
Sample3_3_8	816.4	59.3	87	13.76	8.05	8.29	2.9%
Sample3_3_9	482.8	49.6	90	9.73	6.58	5.92	11.2%

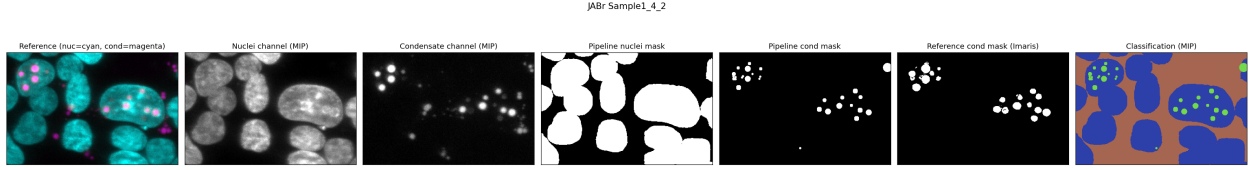
Summary: $n = 28$, mean $|\text{err}| = 13.7\%$, 22/28 within $\pm 20\%$, Pearson $r = 0.942$.

Notable failures (> 25% error)

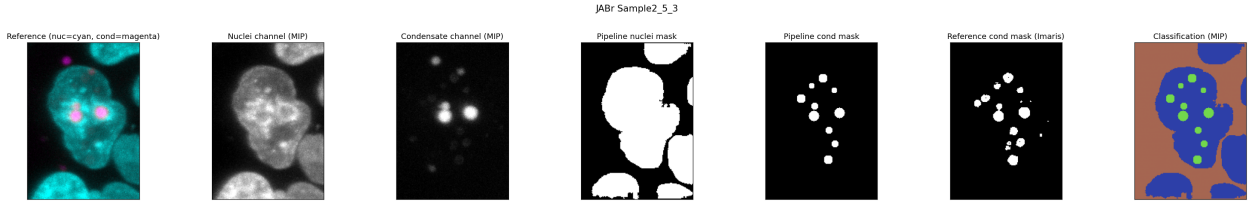
- **Sample1_2_3 (+63%):** smallest reference PC (2.55); the linear calibration floor (intercept ~ 3.0) dominates, so the percentage error is large though the absolute error is only ~ 1.6 PC units.
- **Sample1_1_1 (+43%):** a poor nucleus mask drifting into a dim region inflates the cond/dilute ratio; both `blob_log` and `Cellpose` fail here.
- **Sample3_3_3 (+29%):** an over-bright blob set keeps cond density high relative to the reference PC; not fixed by the gate.

4.4 Visual panels (representative)

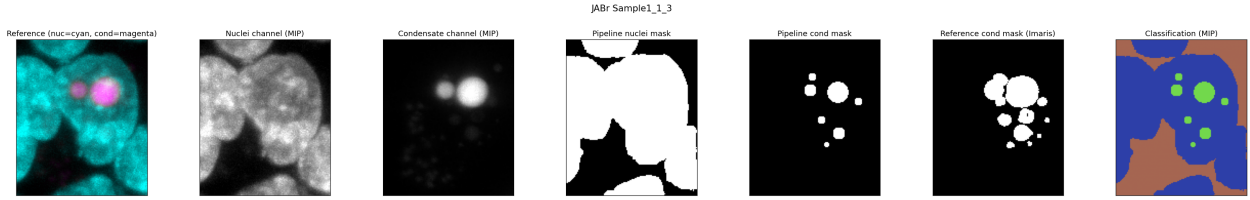
Per-ROI max-intensity Z-projection panels: merged reference (nuclei/condensate) | nuclei channel | condensate channel | pipeline nuclei mask | pipeline condensate mask | reference (Imaris) condensate mask | classification overlay. A representative subset is shown; the full set of 28 panels is in the project repository.



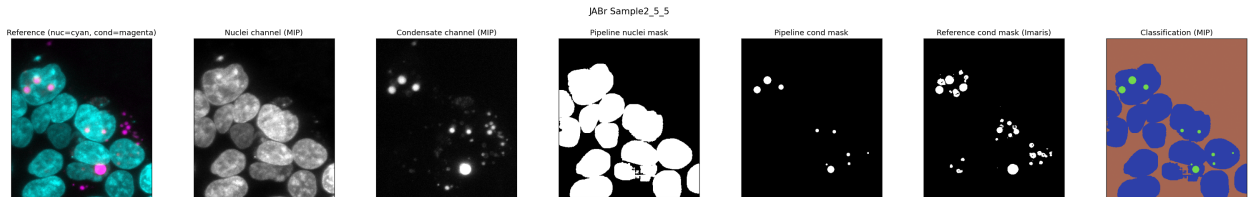
Sample1_4_2 — best case: ref PC 5.05, calibrated 5.06, err 0.3%



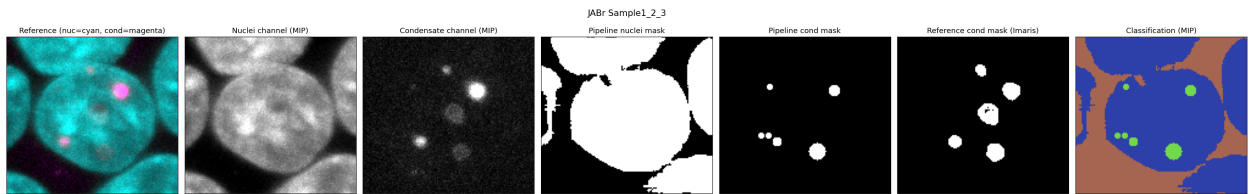
Sample2_5_3 — bundled sample: ref PC 4.56, calibrated 4.67, err 2.4%



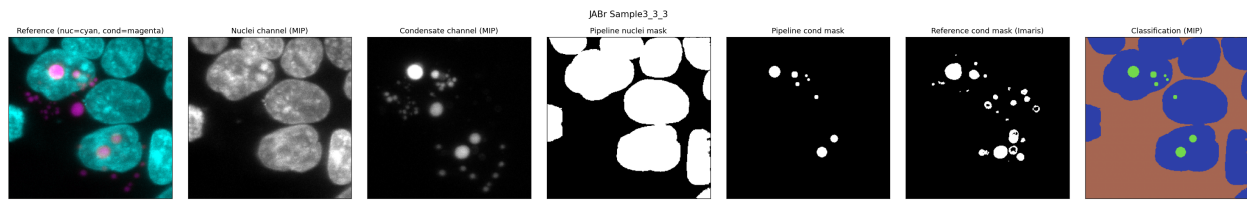
Sample1_1_3 — high PC: ref 19.77, calibrated 19.53, err 1.2%



Sample2_5_5 — rescued by the sphere-overlap gate: was +58%, now ref 6.73 / calibrated 5.72, err 15%



Sample1_2_3 — failure: ref 2.55, calibrated 4.15, err 63% (calibration-floor effect)



Sample3_3_3 — failure: ref 8.82, calibrated 11.42, err 29% (over-bright blobs)